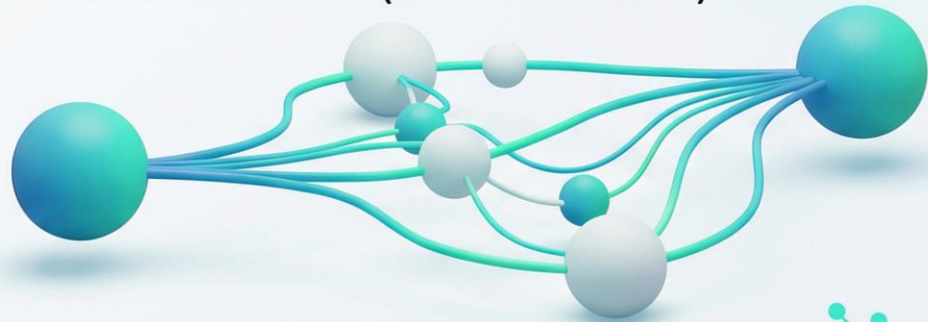


IMPACT 3.0 Proposal

AI-Driven Intelligent Collaborative System for Energy Optimization & ESG Intelligence

MVP to Product-Market Fit Acceleration (TRL-4 → TRL-7)



Submitted by Inspironics Corporation

In collaboration with IIT Palakkad / IPTIF ecosystem

Problem Statement: Energy & ESG Intelligence Gap

Problem Identified



Energy systems are digitized but not collaboratively intelligent



ESG reporting is manual, delayed, compliance-driven



AI exists in silos; no closed-loop optimization

National Relevance



Energy efficiency



Carbon accountability



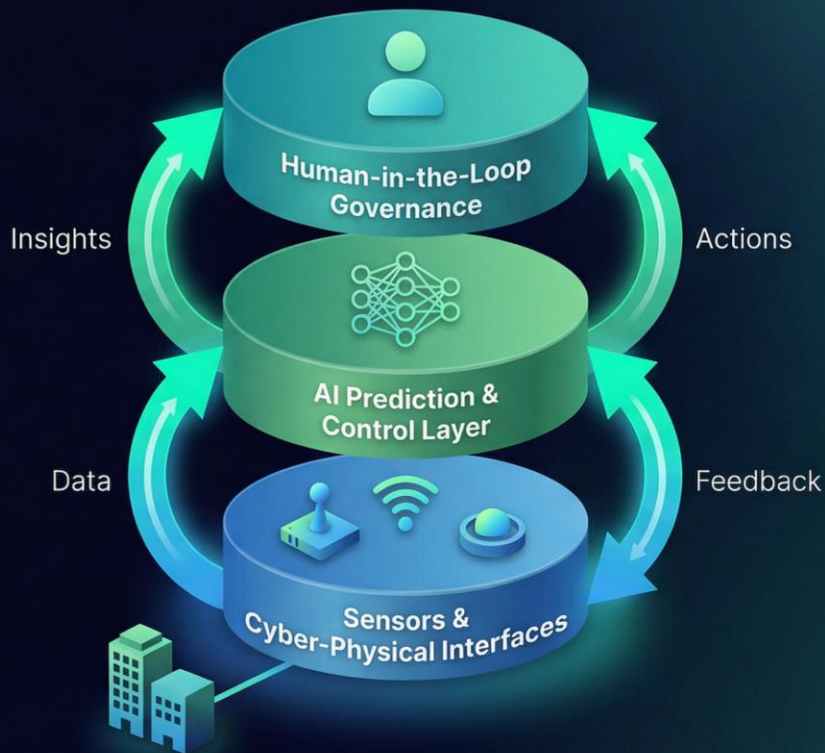
Infrastructure safety & sustainability



Proposed Solution: Intelligent Collaborative System (ICS)

Solution Overview

- AI-driven Intelligent Collaborative System
- Real-time energy sensing + analytics + control
- Closed-loop optimization with ESG intelligence



Core Development Focus: Just-In-Time Scope

FOCUSED DEVELOPMENT AREA (WHY IPTIF)



AI CONTROL & OPTIMIZATION ENGINE

Implementing advanced AI algorithms for real-time energy flow management and predictive analysis.



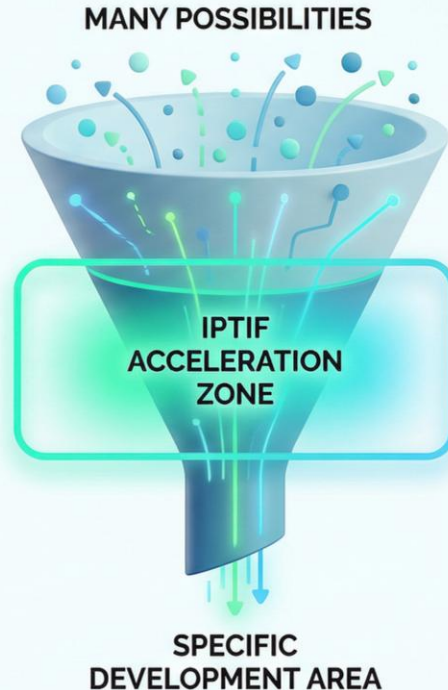
DIGITAL TWIN-BASED ENERGY SIMULATION

Creating dynamic digital models for testing and optimizing energy systems without physical impact.



ESG DATA NORMALIZATION & COMPLIANCE LOGIC

Standardizing environmental, social, and governance data for streamlined reporting and regulatory adherence.



WHY NOW



STRONG MVP EXISTS

A robust minimum viable product is already functional, demonstrating core capabilities.



DEVELOPMENT SLOWED DUE TO CAPITAL & RESEARCH BANDWIDTH CONSTRAINTS

Progress hampered by limited funding and available research resources, requiring strategic acceleration.

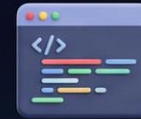
Current Status: Entry TRL-4/5

We Are Here



Entry TRL: 4-5 (Validated components in controlled environment)

What Exists Today



MVP software platform



Initial energy analytics models



Early customer discovery feedback

Target Outcomes: End TRL-7

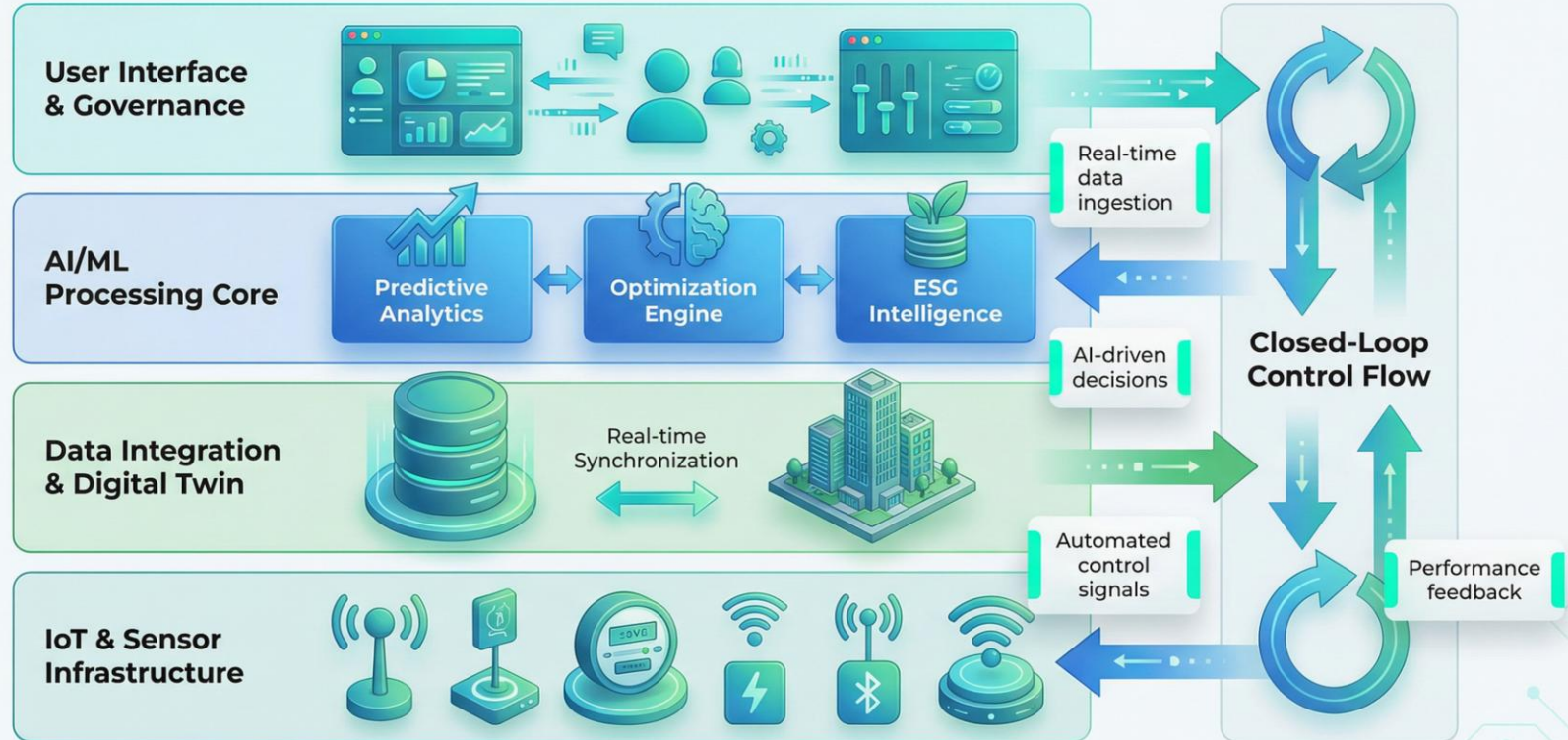


End of Project Deliverables



End TRL: ≥ 7 (Product validated in real user environments)

System Architecture Overview: ICS Platform



Role of IPTIF & IIT Palakkad



Value from IMPACT Program



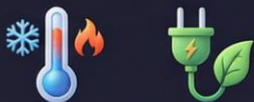
Beyond funding—ecosystem de-risks technical and validation roadmap

Pilot Scenarios & Validation Strategy



Commercial Building

- HVAC optimization
- Occupancy-based control



Hospitality Facility

- Real-time ESG tracking
- Guest comfort + efficiency

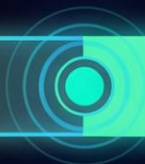


Industrial Campus

- Multi-asset coordination
- Compliance automation



Lab Testing



Controlled Pilot



Field Validation



Performance
Benchmarking

Market & Commercial Path: Parallel Track

Parallel Market Progress



Hospitality & Commercial Buildings



Energy-Intensive Facilities



ESG-Mandated Asset Owners



R&D Track (IMPACT Program)

Prototype

Testing

Refinement

Market Launch

Market Analysis

Pilot Discussions

Partnership Building

Commercial Track

Post-IMPACT Launch



Commercial Launch

Q4 2025



Enterprise Pilots

5-10 sites



Revenue Expansion



Team & Execution Capability



Experience Multi-Disciplinary Pillars

Capability

Experience Multi-Disciplinary Team, systems & engineering, product leadership

Proven engineering leadership

Proven Product Architecture

Proven Product Architecture

Proven Product Architecture

Existing platform architecture firm.

Existing reduces execution risk

AI/ML Development, advisors technology partners, pilot hosts



Strong Ecosystem Support

Energy

Existing platform architecture,

Energy Domain, pilot Domain

IoT integration, technology partners, pilot hosts

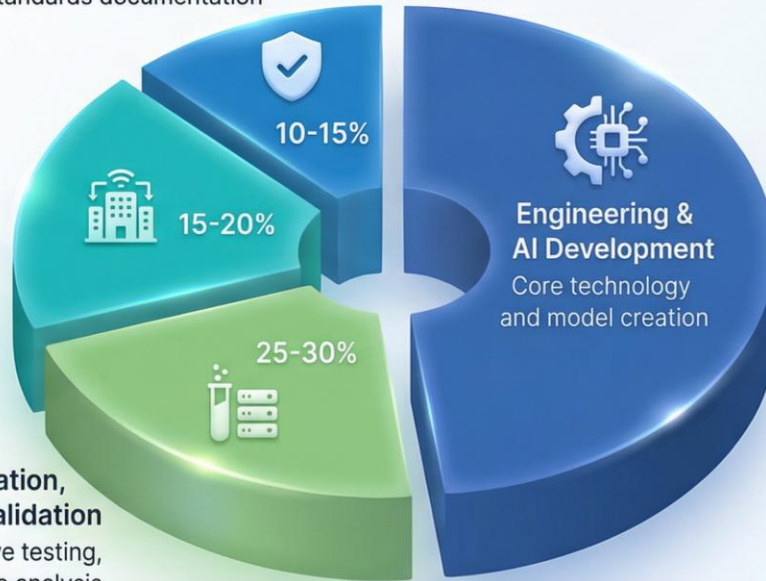


Funding Utilization: ₹50 Lakhs

Compliance & Certification Readiness
Regulatory adherence, standards documentation

Pilot Deployments
Real-world integration,
user feedback

**Simulation,
Testing, Validation**
Iterative testing,
performance analysis



Engineering & AI Development: 45-50%

- Platform development & integration
- AI model training & refinement



Simulation, Testing, Validation: 25-30%

- Lab setup & testing protocols
- Validation with real-world data



Pilot Deployments: 15-20%

- Strategic deployment sites
- User acceptance testing

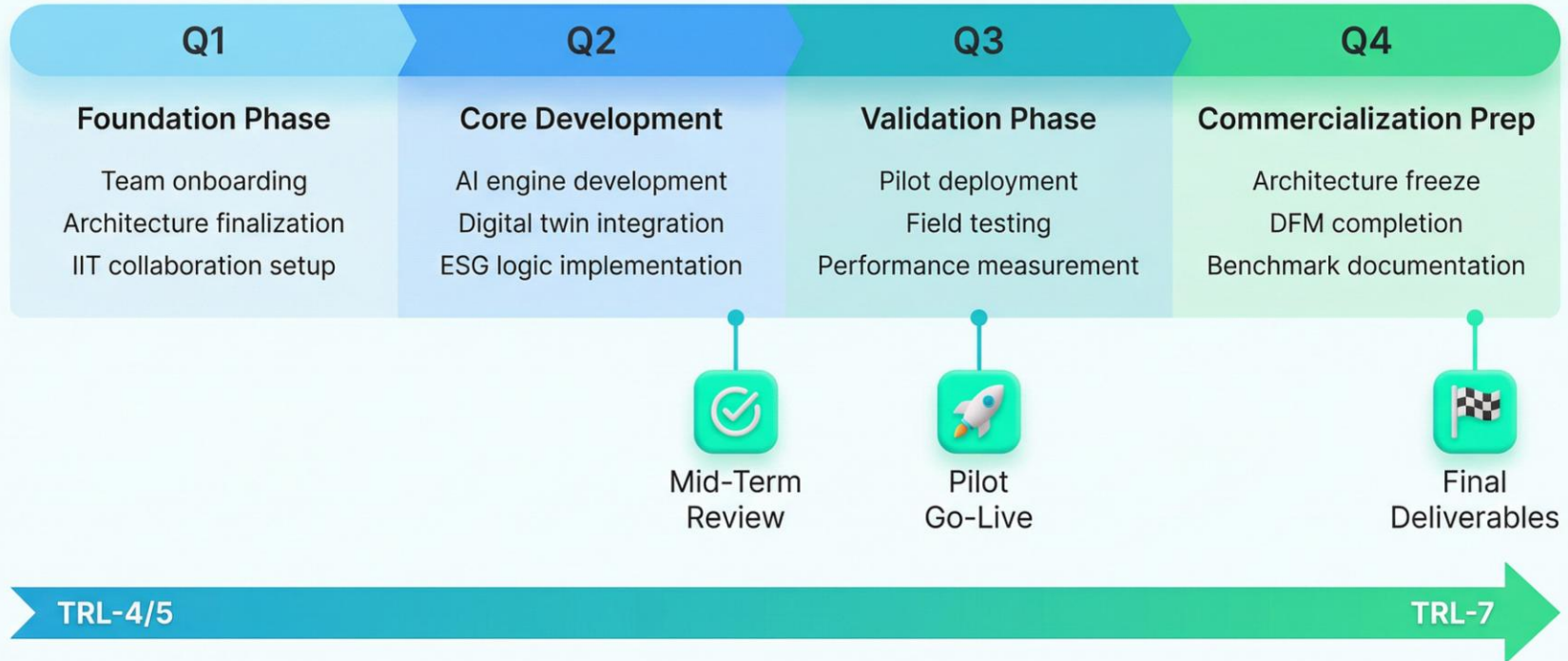


Compliance & Certification Readiness: 10-15%

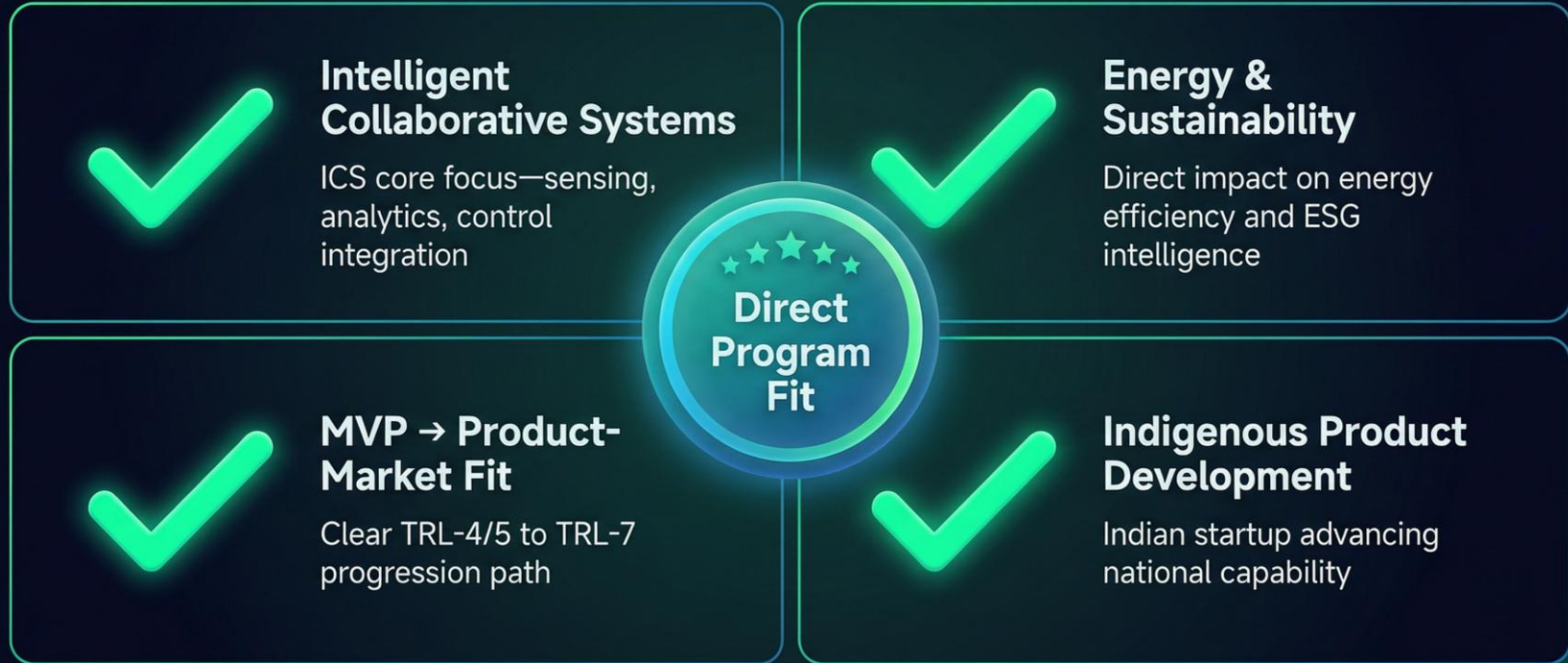
- Standards alignment
- Documentation & certification

*Aligned to IPTIF funding guidelines and **Category 2** requirements*

Timeline & Key Milestones: 12-Month Program



Alignment with IMPACT 3.0 Objectives



Proposal maps directly to IMPACT 3.0 evaluation criteria and program objectives

IMPACT 3.0: Strategic Acceleration Opportunity



Strategic, Focused, Execution-Ready

Constrained scope, high-impact development track



High National Relevance

Energy efficiency, ESG, infrastructure sustainability



Clear Commercialization Path

Immediate market transition post-program

IMPACT 3.0 provides the exact leverage required at this stage—accelerating development while positioning the product for scalable impact.